

Accessible Cooking Tool for Individuals with Visual and Fine Motor Impairments

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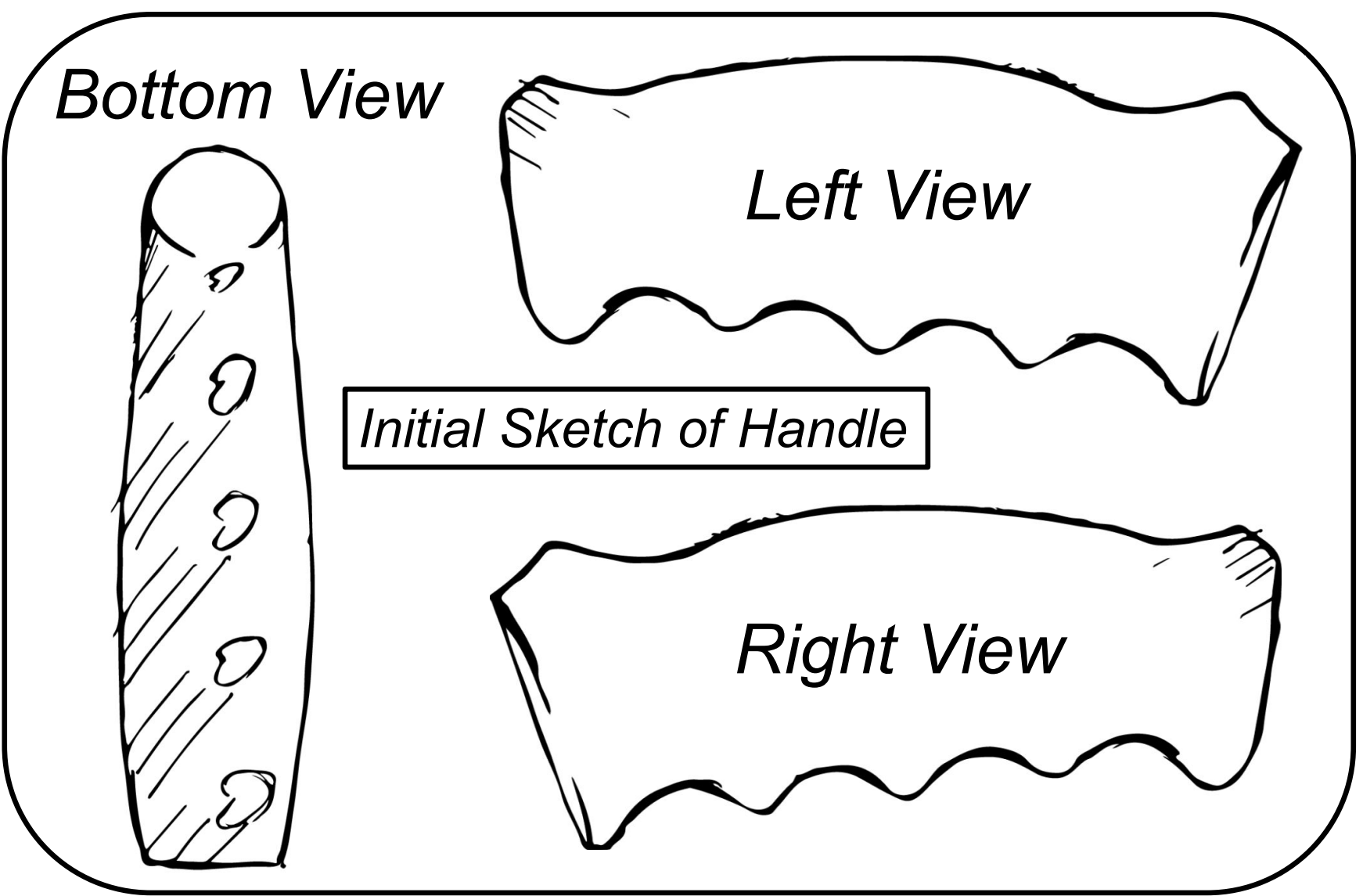
Introduction

Disabilities can make everyday activities, such as eating, running, and working, more challenging for the person. Our co-designer has clinical low vision and fine motor impairments, requiring assistance with certain tasks he performs in his restaurant job, like measuring oil into a tablespoon before pouring it into a bowl. For our project, we plan to solve this problem by using sensors and speakers in our device to hopefully get around any problems that may be caused through blindness and fine motor disabilities.

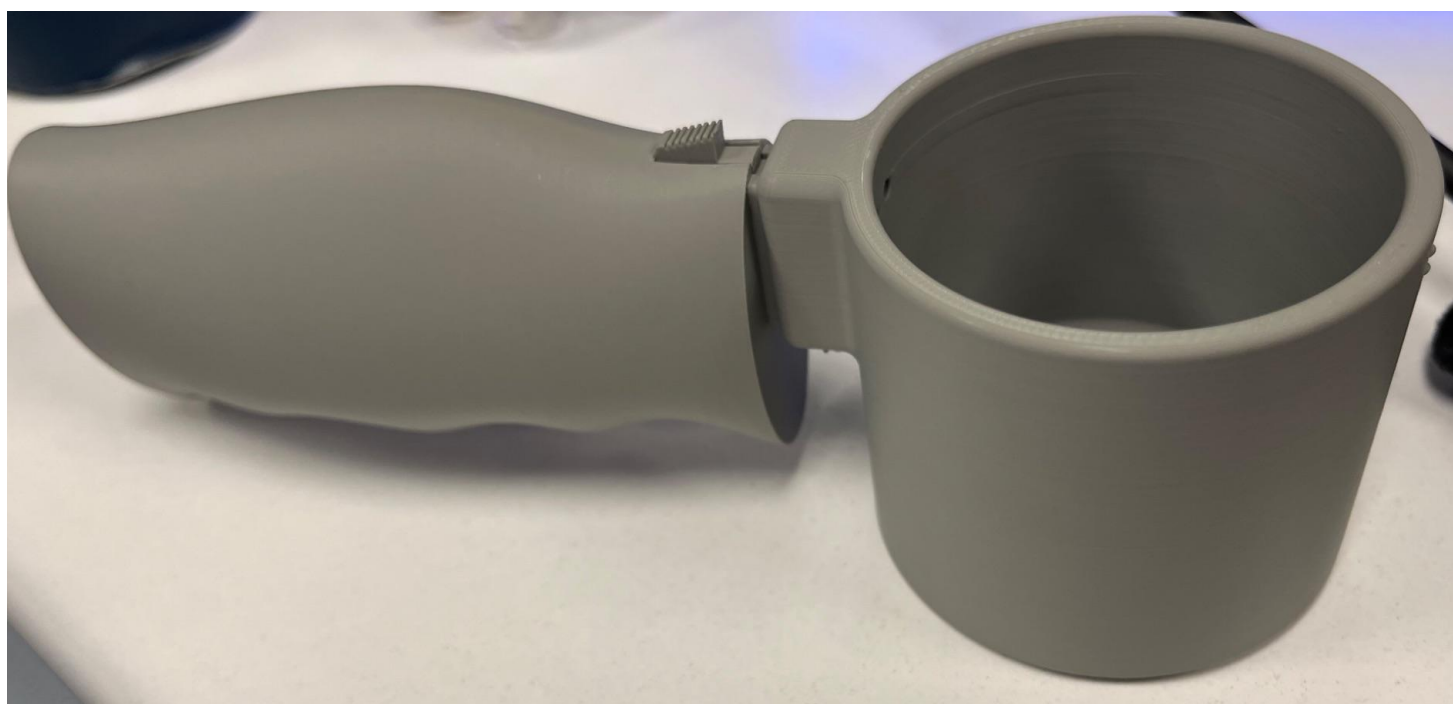
Research Goal

Our goal is to create a tool that helps assist an individual with vision and fine motor impairments in need with pouring materials into containers when cooking using materials like a time of flight sensor, attachable speaker, and programmable circuit boards.

Prototype

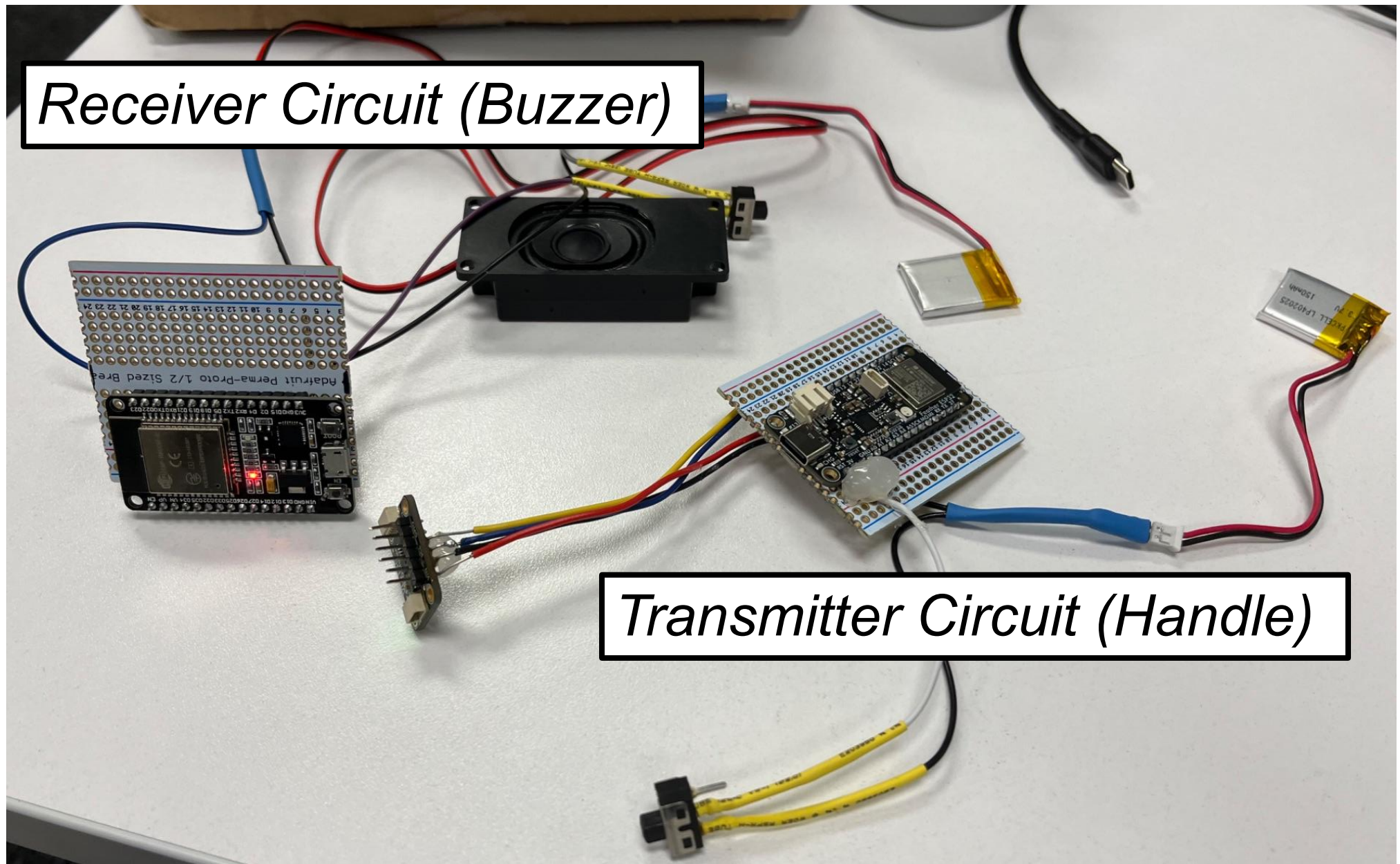


- The image above shows an early sketch of what the handle should look like. We wanted the bottom of the handle to have big humps so that the user can easily get a grip on the device.
- The bottom image shows the initial prototype of the tool. The handle was made hollow so that it could fit the ESP32 board inside. In addition, the cup was retractable, meaning that it could easily come off the handle.



Initial Prototype of Device

Methods



Part 1: Electrical - We created two separate circuits: one for the buzzer, and the other for the handle of the tool. The inside of the handle of the tool contains a time of flight sensor, which calculates the distance of an object by emitting a light signal and measuring the time it takes for the reflection to bounce back. When detecting anything within its range, the sensor sends a message to the buzzer circuit to sound an alarm from the speaker, which then signals the user that it's time to pour down the contents.

Part 2: Programming - Using Arduino IDE, we wrote functional code for both the buzzer and the handle modules so that both would perform their respective jobs. Coding their functions allowed us to manipulate specific details, most notably the range of the sensor.

```
// Target Broadcast Address to avoid missing the D0IT board
uint8_t broadcastMAC[] = {0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF};

// This flags an alert if an object gets closer than 70mm.
if (range > 5 && range < 70) {
  packet.objectDetected = true;
  Serial.println("Object Detected!");
}
```

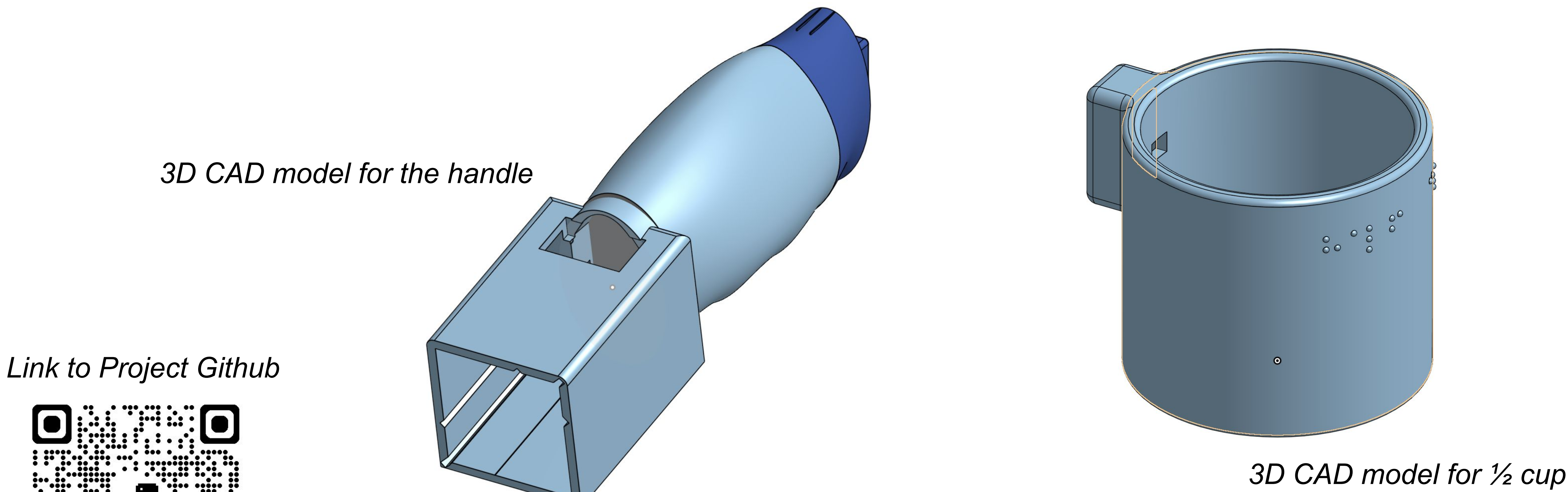
Code shows how handle locates buzzer

```
if (packet.objectDetected) {
  Serial.println("Alarm received! Playing audio alert...");

  // Play dual-beep alert pattern
  tone(SPEAKER_PIN, BUZZ_FREQ, 150);
  delay(200);
  tone(SPEAKER_PIN, BUZZ_FREQ, 150);
} else {
  // Keep speaker quiet if nothing is near the sensor
  noTone(SPEAKER_PIN);
}
```

Code that controls when the buzzer makes sound and when it stays silent

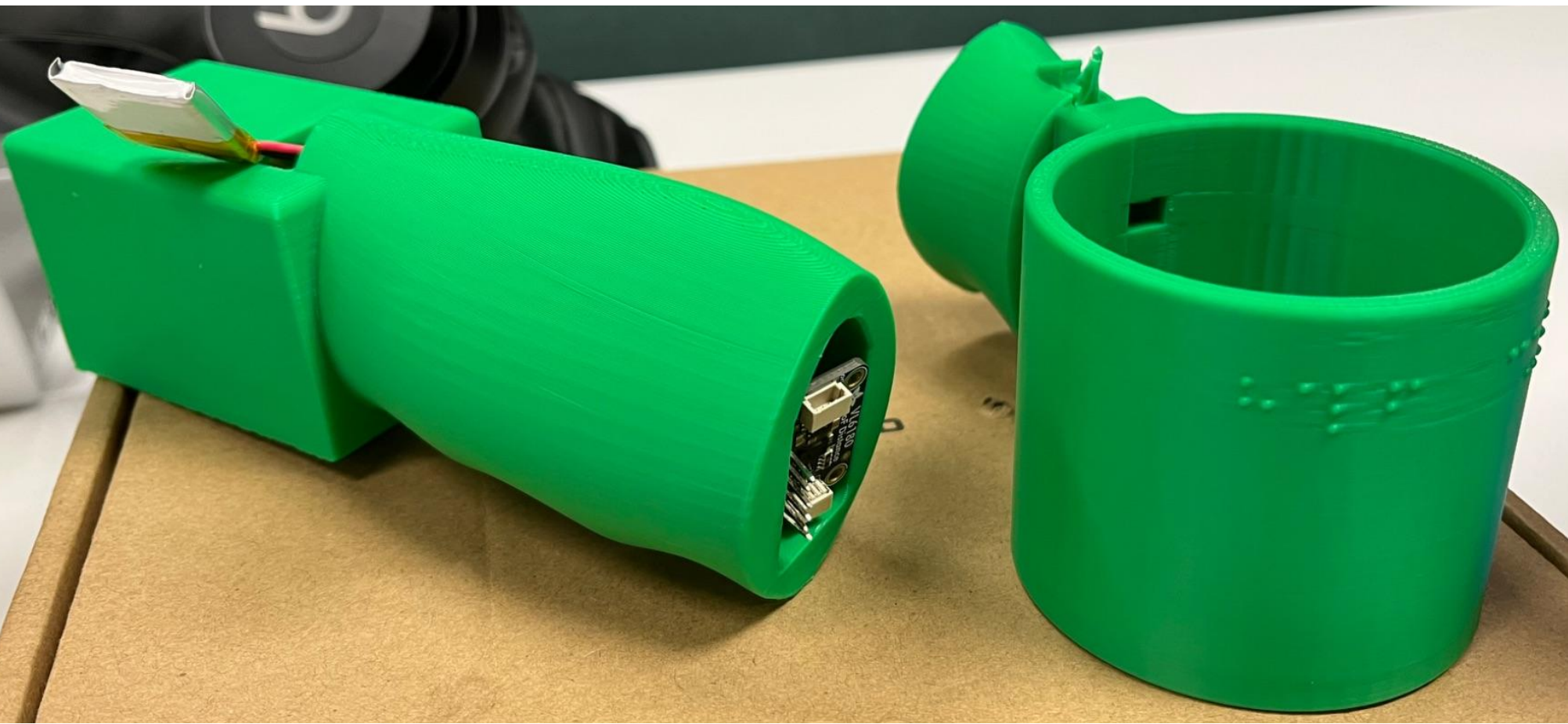
Part 3: Modeling - Seeing how Ernie struggled to hold a small teaspoon, we made the handle for our tool easier to hold for our co-designer so that his fine motor impairments would be less of a problem when trying to grip the tool. For the cup, we added braille on the front of the model so that Ernie can tell which cup he is using despite his low vision.



Link to Project Github



Conclusion



Redesigned Handle /w circuit and 1/2 Cup

- We added a box at the end of the handle so that the entire circuit module could fit inside rather than just the ESP32. Although this solved a major problem, we intend to continue developing the model, as there are still design errors needed to be fixed.
- We found that by making fluid measurements rely on auditory cues rather than visual cues, we can help individuals with visual impairments more accurately determine the correct amount of fluid.
- Our main takeaway was that the co-designer's preferences were just as important as the model's functionality during the design process.

Acknowledgments

Thanks to the following individuals for helping us with the project and providing us with all the support we needed to make it to where we are now.

- Patrick Benfield
- Hello Maker Studio Staff

References

Zhang, X. (2016). Cook blind: Enhancing cooking experiences for visually-impaired people (Order No. 10181152). Available from ProQuest Dissertations & Theses Global. (1850913255). <https://ezproxy.lib.utexas.edu/login?url=https://www.proquest.com/dissertations-theses/cook-blind-enhancing-cooking-experiences-visually/docview/1850913255/se-2>

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